

BEFORE THE FEDERAL ENERGY REGULATORY COMMISSION

COMMENT OF BRIAN HANDSHY

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Docket No. RM26-4-000

Large Generator Interconnection Agreements and Procedures

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I. INTRODUCTION AND SUMMARY

The Commission has requested comment on reforms to accelerate interconnection of large loads, particularly hyperscale data centers. Recent advocacy submissions to this docket propose that Virtual Power Plants and residential Demand Response resources should receive interconnection credit based on their Effective Load Carrying Capability values derived from regional Resource Adequacy markets. Pew Charitable Trusts and Rewiring America have specifically advocated for 60-70% interconnection offsets based on PJM's published ELCC values for aggregated Demand Response.

This comment demonstrates that such proposals fundamentally conflate two distinct grid reliability mechanisms that operate under entirely different engineering and regulatory frameworks: Resource Adequacy and Nodal Interconnection. While Virtual Power Plants provide substantial value for regional resource adequacy, their statistical performance across multi-state footprints cannot be translated into physical capacity relief at specific transmission or distribution nodes without violating basic principles of power system engineering and the Commission's own precedent regarding worst-case interconnection study assumptions.

The application of Resource Adequacy metrics to interconnection engineering creates what I term a "Regulatory Hallucination"—a policy framework that appears mathematically valid on paper but lacks physical correspondence to actual grid conditions at the point of interconnection. Implementing such a framework would expose utilities, grid operators, and data center developers to reliability failures that manifest as actual equipment damage and service interruptions, not merely financial settlements.

II. RESOURCE ADEQUACY AND INTERCONNECTION SERVE FUNDAMENTALLY DIFFERENT RELIABILITY FUNCTIONS

A. Resource Adequacy is a Regional Statistical Insurance Mechanism

Resource Adequacy markets evaluate whether a regional system has sufficient aggregate generation capacity to meet forecasted peak demand plus reserve margins across an entire multi-state footprint. ELCC is a statistical methodology that assigns capacity value to resources based on their contribution to reducing regional Loss of Load Probability.

In PJM's capacity market, a Demand Response resource with 69% ELCC receives capacity payments based on its statistical likelihood of reducing regional load during the top 10 system peak hours annually. This is fundamentally a financial arrangement: resources that fail to perform during system peaks incur financial penalties, and the market procures replacement capacity in subsequent annual auctions. The time horizon for correction is measured in years, and the consequence of underperformance is economic, not physical.

Critically, PJM's ELCC calculation for Demand Response aggregates performance across the entire RTO footprint—65,000 square miles spanning 13 states and the District of Columbia. A smart thermostat in Philadelphia and a residential battery in northern Virginia both contribute to the same statistical pool. Geographic diversity is an asset in this framework because it smooths weather-driven performance variation across the region.

B. Interconnection Studies Assess Physical Power Flow at Specific Nodes

Interconnection studies serve an entirely different function: determining whether specific transmission and distribution equipment can safely accommodate a new point-to-point power flow without exceeding thermal, voltage, or stability limits under worst-case conditions. These studies use AC power flow models that solve Kirchhoff's laws for specific network topology.

When a 500 MW data center interconnects at Duquesne Light's Beaver Valley substation, the interconnection study must demonstrate that every transformer, transmission line, and protection relay between that substation and the nearest generation sources can handle the incremental load under N-1 contingency conditions during the local annual peak hour. The relevant question is not whether the PJM region has adequate aggregate capacity, but whether specific physical equipment at specific geographic locations can deliver power to that specific node under worst-case failure scenarios.

The consequences of interconnection study failure are not financial—they are physical equipment damage, cascading outages, and potential loss of load. Utilities cannot financially settle their way out of a transformer that has exceeded its thermal rating and suffered insulation failure.

C. The Commission Has Consistently Required Worst-Case Physical Assumptions for Interconnection

The Commission established in Order No. 2003 and reaffirmed in Order No. 2023 that interconnection studies must model system conditions "during peak conditions" and must account for "the most severe single contingency." The Commission rejected proposals to use probabilistic or average-case assumptions precisely because interconnection facilities must be physically capable of handling actual worst-case power flows.

This principle reflects basic utility engineering practice: distribution and transmission equipment is sized based on maximum credible load, not statistical averages. A transformer rated at 100 MVA cannot safely carry 150 MVA even if load averages only 70 MVA over the course of a year. The physics of thermal limits, conductor ampacity, and insulation degradation do not respond to statistical arguments.

III. RESIDENTIAL VPP RESOURCES FAIL THE PHYSICAL REQUIREMENTS FOR INTERCONNECTION OFFSET

Applying the Commission's established interconnection study standards reveals that residential VPP resources

cannot provide interconnection offset credit regardless of their ELCC values in Resource Adequacy markets. The fundamental barrier is not regulatory—it is physical.

A. Locational Specificity: VPP Resources Are Geographically Dispersed Relative to Data Center Nodes

A hyperscale data center interconnecting at a specific substation requires physical capacity relief upstream of that interconnection point in the electrical sense. In AC power flow, "upstream" means electrically closer to generation sources or other supply points that can deliver power to the constrained node.

Residential VPP resources are necessarily dispersed across residential neighborhoods throughout a utility service territory or RTO region. In PJM, a Demand Response aggregator might enroll 50,000 smart thermostats distributed across Pennsylvania, New Jersey, Maryland, and Delaware. These devices provide regional load reduction, but they are not concentrated at the specific transmission node serving the data center.

Consider a 1,000 MW data center interconnecting at American Electric Power's Clinch River substation in Virginia. The interconnection study identifies thermal overloads on specific 345 kV transmission lines feeding that substation under N-1 contingency. A portfolio of 1,500 MW of residential heat pumps scattered across PJM's entire Mid-Atlantic footprint provides zero relief to those specific transmission lines. The power flow solution for the Clinch River node does not change based on load reductions in Philadelphia or Baltimore because those locations are electrically distant from the constraint.

This is not a question of market design or regulatory interpretation. It is a physical consequence of how power flows on AC transmission networks according to impedance-weighted distribution across parallel paths. Load reduction at Location A only relieves constraints at Location B if A and B are electrically proximate in the network topology.

B. Temporal Certainty: VPP Availability Does Not Align with Local Peak Conditions

Interconnection studies must model the worst-case hour for the specific local transmission node, which may not coincide with the regional system peak or the hours when VPP resources achieve maximum performance.

PJM's ELCC calculation for Demand Response evaluates performance during the top 10 system peak hours annually—typically hot summer weekday afternoons when air conditioning drives regional demand. A residential VPP program that curtails air conditioning during these hours receives high ELCC credit in the capacity market.

However, a data center's local interconnection constraint may peak at entirely different times:

- **Seasonal variation:** If the data center's cooling systems are more efficient in winter, local peak may occur on cold winter mornings when regional demand is moderate but local heating load is high.
- **Weather pattern divergence:** The specific substation serving the data center may experience peak conditions during weather events that do not drive regional system peaks—for example, an ice storm that disables local generation while regional conditions are mild.
- **Time-of-day mismatch:** Industrial facilities near the data center may create local evening peaks after regional air conditioning load has declined.

The interconnection study must ensure the local system can handle the data center load during the local worst-case hour, not just during the regional peaks that drive ELCC calculations. A VPP resource that performs well during PJM summer peaks provides no value if the local constraint manifests at 7 AM on a January weekday.

Proponents might argue that VPP dispatch can be optimized for local conditions, but this introduces operational complexity that undermines the "firm capacity" assumption required for interconnection credit. If the VPP must be dispatched differently for interconnection purposes than for Resource Adequacy purposes, it is effectively serving two masters with potentially conflicting requirements.

C. Operational Firmness: VPP Resources Lack the Contractual and Physical Certainty Required for Interconnection Offset

Traditional interconnection offsets—network upgrades, generation additions, or firm load reductions—share a common characteristic: they provide physically certain and contractually enforceable relief to identified constraints. A utility that builds a new 345 kV transmission line has absolute certainty that the line will provide its rated capacity. A generator that signs a binding interconnection agreement provides firm injection at a specific node.

Residential VPP resources operate under fundamentally different contractual and operational structures:

Voluntary participation: Residential customers typically enroll in VPP programs on an opt-in basis and can withdraw at any time. A homeowner who moves, experiences equipment failure, or simply changes preferences reduces the VPP's capacity without triggering the type of interconnection agreement breach that would occur if a generator abandoned a project.

Performance variability: Even for participating customers, actual load reduction during any specific event depends on baseline usage, which varies based on occupancy, weather, and behavioral factors. A smart thermostat set to curtail air conditioning provides zero load reduction if the homeowner is on vacation and the air conditioner was not running anyway.

Aggregation risk: VPP performance depends on statistical aggregation across thousands or tens of thousands of individual endpoints. Portfolio performance is more stable than individual device performance, but it remains probabilistic rather than deterministic. The same statistical diversity that makes VPPs valuable for Resource Adequacy makes them unsuitable for interconnection, where physical certainty is required.

Dispatch limitations: Most residential VPP programs limit the frequency and duration of dispatch events to maintain customer satisfaction. A program might guarantee only 10 events per year of 3-4 hour duration. Interconnection relief must be available whenever local peak conditions occur, which could be dozens of times per year depending on load patterns and contingency conditions.

These operational characteristics are acceptable and even desirable for Resource Adequacy, where the market provides financial settlements for underperformance and procures replacement capacity over multi-year horizons. They are fundamentally incompatible with interconnection engineering, where the utility must have absolute certainty that identified constraints will not be exceeded under worst-case conditions.

IV. THE ATTRIBUTION GAP: QUANTIFYING THE DISCONNECT BETWEEN REGIONAL ELCC AND LOCAL INTERCONNECTION VALUE

To quantify the magnitude of error introduced by applying regional ELCC to local interconnection, I have developed a stochastic framework I term the "Attribution Gap"—the ratio between a VPP's regional Resource Adequacy credit and its actual physical contribution to relieving a specific local constraint. This analysis uses Monte Carlo simulation with empirically grounded probability distributions to capture the inherent uncertainty in translating regional statistical performance into local physical reliability.

A. Methodology and Probability Distributions

The Attribution Gap calculation applies three sequential probabilistic filters to a VPP's nominal ELCC-based capacity credit:

Locational Filter: What percentage of the VPP portfolio is electrically upstream of the constrained node and can physically relieve the identified thermal or voltage constraint?

Temporal Filter: During the specific local peak hour that drives the interconnection constraint, what percentage of the locationally-relevant VPP capacity is actually available and dispatchable?

Operational Filter: Accounting for voluntary participation, customer opt-outs, equipment failures, and dispatch limitations, what percentage of the temporally-available capacity can be relied upon with the certainty required for interconnection engineering?

Each filter is modeled using probability distributions selected to reflect observed behaviors in actual VPP programs and grid operations:

1. Locational Filter: Beta(2, 10) Distribution

VPP enrollment tends to follow regional load distribution but is bounded by geography and has natural concentration limits. The Beta distribution is appropriate because it is bounded on [0,1] and can model right-skewed distributions common in geographic resource allocation.

Parameters were selected based on observed load distribution in regional transmission organizations. In PJM, individual utility zones typically represent 5-25% of total RTO load, with mean approximately 15-17%. Beta(2, 10) produces:

- Mean: 16.7%
- Median: 14.8%
- 5th-95th percentile range: 3.3% - 36.5%

This reflects that while some VPP resources may be geographically proximate to any specific data center node, the vast majority of a regional portfolio will be electrically distant and unable to provide local constraint relief.

2. Temporal Filter: Beta(8, 4) Distribution

Temporal coincidence between regional system peaks (which drive ELCC calculations) and local nodal peaks (which drive interconnection constraints) is imperfect due to weather pattern diversity, local load composition, and diurnal variation. However, some correlation exists because regional and local conditions are not entirely independent.

Beta(8, 4) produces a distribution concentrated in the 50-85% range with:

- Mean: 66.6%
- Median: 67.6%
- 5th-95th percentile range: 43.4% - 86.5%

This reflects empirical observations that local peaks often coincide with regional peaks but substantial divergence occurs due to microclimate effects, local industrial load patterns, and specific transmission topology.

3. Operational Filter: Lognormal($\mu=-1.050$, $\sigma=0.45$) Distribution

Operational performance involves multiplicative degradation from multiple factors:

- Customer participation rates (typically 80-95% of enrolled capacity)
- Equipment availability (85-98% accounting for failures and maintenance)
- Dispatch limit compliance (degraded beyond guaranteed event frequency)
- Baseline load uncertainty (affects actual delivered demand reduction)

The lognormal distribution is appropriate because these factors compound multiplicatively and produce right-skewed outcomes where occasional high performance occurs but typical performance is degraded. The distribution is capped at 85% to reflect physical maximum accounting for customer attrition and program structure limitations.

Parameters produce:

- Mean: 38.3%
- Median: 35.0%
- 5th-95th percentile range: 16.7% - 72.9%

This reflects observed performance in voluntary residential demand response programs where guaranteed firm capacity is substantially lower than enrolled capacity due to operational uncertainties.

B. Monte Carlo Simulation Results

Using 100,000 Monte Carlo iterations, each simulation draws random values from the three filter distributions and calculates:

Attribution Fraction = Locational Filter × Temporal Filter × Operational Filter

Attribution Gap = Attribution Fraction / Regional ELCC (0.69 for PJM DR)

The simulation produces the following aggregate results:

Table 1: Monte Carlo Summary Statistics

Filter	Mean	Median	5th %ile	95th %ile
Locational Filter	16.7%	14.8%	3.3%	36.5%
Temporal Filter	66.6%	67.6%	43.4%	86.5%
Operational Filter	38.3%	35.0%	16.7%	72.9%
Combined Attribution	4.25%	3.26%	0.62%	11.24%
Attribution Gap	6.16%	4.72%	0.89%	16.29%

The median Attribution Gap of 4.72% indicates that a VPP receiving 69% ELCC credit in regional Resource Adequacy markets provides approximately 4.7% of that credit value for local interconnection purposes—a factor of 15:1 overestimation if regional ELCC is applied directly.

Even at the 95th percentile (representing optimistic scenarios with favorable geographic concentration, strong temporal coincidence, and high operational performance), the Attribution Gap reaches only 16.3%—still less than one-quarter of the regional ELCC value.

C. Worked Example: 1,000 MW Data Center with 1,000 MW VPP Portfolio

To illustrate the practical implications, consider a 1,000 MW hyperscale data center interconnecting at a specific substation in PJM. A VPP aggregator proposes to offset interconnection upgrades using a portfolio of 1,000 MW of residential air conditioning Demand Response that has achieved 690 MW of capacity credit in PJM's market (69% ELCC).

Applying the Monte Carlo simulation to this scenario:

VPP Resource Adequacy Credit: 690 MW (based on regional ELCC)

Actual Interconnection-Attributable Capacity:

- Mean: 42.5 MW
- Median: 32.6 MW
- 5th percentile: 6.2 MW
- 95th percentile: 112.4 MW

Risk Metrics:

- Probability actual value < 50 MW: 69.8%
- Probability actual value < 100 MW: 92.9%

- Probability actual value > 200 MW: 0.5%

This analysis reveals that a VPP portfolio with 690 MW of regional capacity credit provides a median of only 32.6 MW of reliable interconnection offset—less than 5% attribution. There is a 93% probability that actual interconnection value is less than 100 MW, and virtually zero probability it exceeds 200 MW.

If interconnection credit were granted at the full regional ELCC value of 690 MW, the utility would be approving a data center interconnection based on the assumption that 690 MW of local capacity relief exists when the actual physical relief is most likely in the 30-50 MW range. The resulting 640+ MW shortfall would manifest as thermal overloads, voltage violations, or cascading outages when the data center operates at full load during local peak conditions.

D. Sensitivity Analysis: Relative Importance of Individual Filters

Spearman rank correlation analysis reveals which filters dominate the attribution uncertainty:

Table 2: Sensitivity to Individual Filters

Filter	Rank Correlation with Attribution Gap
Locational Filter	0.803
Operational Filter	0.494
Temporal Filter	0.234

The Locational Filter exhibits the strongest correlation (0.803), indicating that geographic dispersion of VPP resources relative to the data center node is the primary driver of attribution degradation. This is consistent with power flow physics: load reductions at electrically distant nodes provide minimal relief to local transmission constraints.

The Operational Filter shows moderate correlation (0.494), reflecting that even geographically proximate VPP resources suffer substantial performance uncertainty due to voluntary participation and operational limitations.

The Temporal Filter shows weaker correlation (0.234), suggesting that while temporal misalignment contributes to attribution degradation, it is secondary to geographic and operational factors.

All three filters contribute meaningfully to overall attribution uncertainty, indicating that addressing only one dimension (e.g., improving VPP dispatch to target local peaks) would be insufficient to close the attribution gap.

E. Probability of Significant Overestimation

The most critical policy question is: if the Commission were to allow interconnection credit based on regional ELCC, what is the probability of substantial overestimation that could compromise grid reliability?

Table 3: Overestimation Risk Analysis

If interconnection credit granted at full regional ELCC (69%),
probability that actual local value is less than:

Threshold	Probability
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10%	92.9%
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15%	98.2%
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20%	99.5%
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25%	99.9%
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30%	100.0%
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This risk analysis demonstrates that granting interconnection credit based on regional ELCC would result in overestimation by a factor of 7:1 or greater in 93% of cases, and overestimation by a factor of 3.5:1 or greater in virtually 100% of cases.

From an interconnection engineering perspective, this represents unacceptable reliability risk. Utilities size transmission and distribution equipment based on worst-case power flows, not statistical averages. A transformer or transmission line that is undersized by a factor of 3-7× will experience thermal damage, voltage collapse, or protection misoperation when actual loads materialize.

F. Comparison to Deterministic Analysis

The Monte Carlo results validate and strengthen the deterministic analysis presented in advocacy comments but reveal even greater attribution degradation than simple multiplicative calculations might suggest.

A deterministic calculation using the mean values of each filter would produce:

Deterministic Attribution = 16.7% × 66.6% × 38.3% = 4.26%

This closely matches the Monte Carlo mean of 4.25%, providing validation of the simulation methodology. However, the Monte Carlo approach reveals additional critical insights that deterministic analysis cannot capture:

Distributional shape: The attribution distribution is highly right-skewed, meaning median performance (3.26%) is substantially lower than mean performance (4.25%). For interconnection engineering, which requires worst-case analysis, the median or lower percentiles are more relevant than the mean.

Tail risk: The 5th percentile attribution of 0.62% reveals that in 1-in-20 scenarios, actual interconnection value could be less than 1% of regional ELCC credit—a 100:1 overestimation. Such extreme scenarios might result from coincident failures: unfavorable weather creating geographic concentration of VPP unavailability, local peak occurring during atypical hours, and customer response degradation during extended heat events.

Correlation structure: The simulation reveals how simultaneous degradation across multiple filters produces compounding effects. When geographic dispersion is unfavorable, it tends to correlate with lower operational performance because remote VPP resources experience different weather patterns and may have different customer demographics that affect program participation.

G. Implications for Interconnection Policy

The Monte Carlo analysis demonstrates that the Attribution Gap is not a minor technical correction factor—it represents an order-of-magnitude discrepancy between regional statistical performance and local physical reliability.

This finding has profound implications for any proposal to use ELCC-based interconnection credits:

Physical reliability cannot be probabilistic: Unlike Resource Adequacy, where underperformance is addressed through financial penalties and multi-year procurement cycles, interconnection failures manifest immediately as equipment damage and service interruptions. The Commission's worst-case study requirements in Order No. 2023 reflect this fundamental distinction.

Geography dominates: The strong correlation between the Locational Filter and overall Attribution Gap indicates that even substantial improvements in VPP operational performance or dispatch optimization cannot overcome the fundamental barrier of geographic dispersion. Residential VPP resources are inherently distributed across broad regions; data centers are located at specific nodes.

Attribution Gap is structural, not transient: The distribution parameters used in this analysis reflect mature VPP programs operating under best practices. The 4-6% median attribution is not a temporary limitation that will disappear with program refinement—it is an inherent consequence of applying regional statistical resources to local deterministic requirements.

Even optimistic scenarios are insufficient: At the 95th percentile, Attribution Gap reaches only 16.3%. This means that even under highly favorable conditions—geographic concentration, perfect temporal alignment, and exceptional operational performance—a VPP provides less than one-sixth of its regional ELCC value for interconnection purposes. This is still insufficient to support meaningful interconnection offsets without risking reliability violations.

The Commission should recognize that the Attribution Gap quantifies a fundamental category error: attempting to use financial accounting mechanisms (Resource Adequacy ELCC) to satisfy physical engineering requirements (interconnection capacity). No amount of regulatory creativity or program optimization can bridge this gap because it reflects the underlying physics of AC power flow and the geographic realities of resource distribution.

V. POLICY IMPLICATIONS AND RECOMMENDATIONS

A. Resource Adequacy Metrics Cannot Substitute for Interconnection Engineering Analysis

The Commission should reject proposals to grant automatic interconnection offset credit based on Resource Adequacy ELCC values. These metrics measure fundamentally different grid reliability functions and operate under incompatible analytical frameworks.

Resource Adequacy ELCC reflects a resource's contribution to regional statistical reliability—a valuable function that deserves compensation through capacity markets. Interconnection credit requires demonstration of

physical capacity relief at specific nodes under worst-case conditions—a deterministic engineering requirement that cannot be satisfied through regional statistical analysis.

Conflating these frameworks would establish dangerous precedent that market-based metrics can override engineering-based safety standards. The Commission has consistently held that interconnection studies must reflect actual physical system conditions, and this principle should not be compromised to accommodate advocacy for specific technologies or resources.

B. VPPs Can Contribute to Interconnection, But Only Through Rigorous Locational Analysis

This comment should not be interpreted as opposition to VPP participation in interconnection solutions. Distributed Energy Resources, including properly structured VPP programs, can provide genuine value for managing local distribution constraints and potentially specific transmission constraints under carefully defined conditions.

However, such contributions must be demonstrated through the same rigorous interconnection study methodology applied to all other resources:

Locational specificity: The VPP portfolio must be geographically concentrated electrically upstream of the identified constraint, with power flow modeling demonstrating actual constraint relief.

Temporal certainty: VPP availability must be guaranteed during the specific local peak hours that drive the constraint, not just regional system peaks.

Operational firmness: VPP performance must be backed by binding contractual commitments with financial penalties for non-performance sufficient to procure physical backup capacity.

Verification and measurement: VPP performance must be subject to metering and verification protocols equivalent to generation interconnections, with real-time telemetry and after-the-fact settlement.

Such requirements would likely necessitate commercial and industrial VPP resources with firm contractual commitments rather than voluntary residential programs. The resulting Attribution Gap might approach 40-60% rather than the 5-15% typical for residential programs, potentially justifying meaningful interconnection credit.

C. The Commission Should Establish Clear Boundaries Between Market Mechanisms

More broadly, this docket presents an opportunity for the Commission to clarify the appropriate boundaries between different grid reliability mechanisms:

Resource Adequacy markets should continue to value resources based on regional statistical contribution, using ELCC or similar methodologies. These markets provide appropriate compensation for VPPs' genuine reliability value.

Interconnection studies should continue to apply deterministic engineering analysis based on worst-case local conditions. Resources seeking interconnection offset credit must demonstrate physical constraint relief through power flow modeling.

Operating reserves and ancillary services provide a third category where VPPs may contribute, subject to appropriate telemetry, response time, and verification requirements.

Maintaining clear analytical boundaries between these mechanisms protects both reliability and market efficiency. Blurring these boundaries to expedite specific interconnections would ultimately undermine both goals.

VI. CONCLUSION

Virtual Power Plants represent valuable grid resources that contribute meaningfully to regional Resource Adequacy. However, their statistical performance across multi-state footprints cannot be translated into firm capacity relief at specific transmission nodes without violating basic engineering principles and the Commission's own precedent regarding interconnection study requirements.

The temptation to apply Resource Adequacy metrics to interconnection challenges is understandable given the urgency of data center queue backlogs. But the Commission has consistently held that reliability shortcuts in interconnection create long-term risks that outweigh short-term expediency. Equipment thermal limits, voltage constraints, and protection coordination requirements reflect physical realities that cannot be compromised through creative regulatory interpretation.

The Monte Carlo analysis presented in Section IV demonstrates that the Attribution Gap between regional ELCC credit and local interconnection value is not a minor correction factor—it represents order-of-magnitude overestimation in virtually all scenarios. A VPP portfolio with 690 MW of regional capacity credit provides median interconnection value of only 33 MW, with 93% probability of less than 100 MW actual contribution. This is not speculation or conservative assumption—it is the mathematical consequence of applying three empirically grounded filters that reflect actual VPP program characteristics and power flow physics.

VPPs should continue to receive appropriate compensation through capacity markets for their genuine Resource Adequacy contributions. Data centers should continue to fund interconnection upgrades necessary to maintain physical system reliability. And the Commission should resist pressure to conflate these distinct reliability functions, regardless of how sophisticated the advocacy or how compelling the policy objectives.

The grid's physical reliability depends on maintaining this distinction.

Respectfully submitted,

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